

Res-GCN: Identification of protein phosphorylation sites using graph convolutional network and residual network

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ABSTRACT

An essential post-translational modification, phosphorylation is intimately related with a wide range of biological activities. The advancement of effective computational methods for correctly recognizing phosphorylation sites is important for in-depth understanding of various physiological phenomena. However, the traditional method of identifying phosphorylation sites experimentally is time-consuming and laborious, which makes it difficult to meet the processing demands of today's big data. This research proposes the use of a novel model, Res-GCN, to recognize the phosphorylation sites of SARS-CoV-2. Firstly, eight feature extraction strategies are utilized to digitize the protein sequence from multiple viewpoints, including amino acid property encodings (AAindex), pseudo-amino acid composition (PseAAC), adapted normal distribution bi-profile Bayes (ANBPB), dipeptide composition (DC), binary encoding (BE), enhanced amino acid composition (EAAC), Word2Vec, and BLOSUM62 matrices. Secondly, elastic net is utilized to eliminate redundant data in the fused matrix. Finally, a combination of graph convolutional network (GCN) and residual network (ResNet) is used to classify the phosphorylated sites and output predictions using a fully connected layer (FC). The performance of Res-GCN is tested by 5-fold cross-validation and independent testing, and excellent results are obtained on S/T and Y datasets. This demonstrates that the Res-GCN model exhibits exceptional predictive performance and generalizability.

1. Introduction

Protein post-translational modification (PTM) is the addition or removal of chemical groups to a translated protein, which can lead to further functional diversity in proteomics. Several processes that control an organism's physiological functions, such as cell biological activities including signal transduction, DNA damage repair, and cell cycle regulation, are regulated by PTM, a common mechanism of organismal genesis. With the study of modern proteomics techniques, it has been found that the diversity of protein modification processes presents in eukaryotic cells, and more than 600 types of PTM have been identified (Khoury et al., 2011), which are crucial to many life activities. Among the most prevalent and significant PTM in cells is protein phosphorylation, which usually enriched on serine (S), threonine (T), and tyrosine

(Y) residues. Phosphorylation has become the preferred modality to regulate many physiological activities in vivo, such as cellular processes of DNA repair, growth, motility, and metabolism (Li et al., 2010; Trost and Kusalik, 2011). Therefore, the prediction and study of protein phosphorylation sites are of theoretical significance for understanding the biological system of proteins and in-depth understanding of various physiological phenomena, and at the same time provide new ideas to guide the design and development of basic drugs.

Although there are a series of research results for protein post-translational modification site prediction, there are still many difficulties and challenges. (1) Protein modification sites are diverse and complex, including different types of modifications (e.g., phosphorylation, succinylation, acetylation, etc.), different modification positions, and modification sequence patterns. This diversity and complexity make

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the design and training of predictive models more difficult. (2) Information about structure and evolution can be found in protein sequences, so only one feature extraction algorithm cannot characterize the corresponding biological functions. (3) An effective feature selection method can boost the model's performance, but some methods directly discard some features that do not contribute much. Some information may be lost if these characteristics are immediately removed, they are not entirely worthless. (4) A variety of different methods and models for protein modification site prediction exist, such as machine-learning-based methods, sequence pattern and structural pattern recognition, etc. It is a challenge to select the appropriate model and evaluate its performance. Therefore, a popular path at the forefront of protein PTM is the creation of novel computational techniques to increase the speed and precision of sites prediction.

To tackle the challenges, this article develops a novel approach to predict two types of protein phosphorylation sites. This study's primary contributions are:

(1) Eight feature extraction methods including AAindex, PseAAC, ANBPB, DC, BE, EAAC, Word2Vec, and BLOSUM62 matrices are utilized to digitize the protein sequence from multiple viewpoints, where the features obtained from different perspectives are fused in parallel into a complex feature vector.

(2) Elastic net is utilized to eliminate redundant data in the fused matrix, with the goal of picking the best features and removing redundant information and irrelevant noise, which effectively solves the bias caused by data redundancy on the results.

(3) Combining graph convolutional network (GCN) and residual network (ResNet) to build a prediction network for classifying SARS-CoV-2 phosphorylation sites. The GCN model, as an input layer of ResNet, can take the information of the neighboring nodes into account, to better model the relationship between nodes and the sequence features.

(4) Both the two-sided T-test and five-fold cross-validation are applied to evaluate the model's ability to predict. Other phosphorylation sites are utilized as well to check the model's generalizability.

There will be four sections in the remaining portion of the article: [Section 2](#) is an introduction to relevant work. [Section 3](#) covers the model's methodology as well as its evaluation method. [Section 4](#) discusses the superiority of the proposed methods in this study. [Section 5](#) outlines the findings and recommendations for future work.

2. Relevant work

Many traditional experimental methods, such as low-throughput 32p-labeling, mass spectrometry, and phospho-specific antibodies, have been employed for determining phosphorylation sites ([Aponte et al., 2009](#); [Beausoleil et al., 2006](#)). However, because traditional methods for identifying PTM are very inconvenient to use and limited by time and equipment, making them publicly available on large-scale datasets is challenging ([Khan et al., 2019](#)). In contrast, computational methods have been rapidly developed, making it relatively easy and fast to identify protein post-translational sites. Identifying phosphorylation sites has been accomplished through the development of more than 40 computational techniques, many of which rely on machine learning algorithms.

[Dou et al. \(2014\)](#) used a support vector machine combined with eight distinct sequence-level scoring functions, and validated the method using ten-fold cross-validation, obtaining AUC scores of 0.8405, 0.8183, and 0.7383 for the common phosphorylation-modifying residues in animals. Developed by [Li et al. \(2018\)](#), Quokka is a logistic regression-based method that predicts protein kinase (PK)-catalyzed protein phosphorylation sites using multiple sequence score functions. [Fan et al. \(2014\)](#) developed a random forest-based prediction model and methodically and hierarchically predicted protein phosphorylation sites from the perspectives of gene ontology (GO) and protein interactions (PPI). A machine learning approach called ScerePhoSite was suggested

by [Wang et al. \(2023\)](#) to locate fungal phosphorylation sites. To provide the optimal subset of features, it executes materialized feature representation of sequence fragments and leverages the significance of LGB features to optimize them through the sequence orthogonal algorithm. A program called PPSP (Prediction of PK-specific Phosphorylation site) was established by [Xue et al. \(2006\)](#) and was in view of the Bayesian Decision Theory (BDT) method. It can accurately predict possible phosphorylation sites in over 70 protein kinases (PKs).

Traditional machine learning methods have made significant improvements, but the existing experimental and machine learning methods face the same problem for the massive amount of protein data information. That is, for many modification sites, the prediction accuracy and efficiency cannot be balanced. With the research on artificial neural networks, deep learning methods with advantages such as high learning capacity and good adaptability are widely used.

[Ahmed et al. \(2021\)](#) built DeepPPSite using a stacked long short-term memory recurrent network. It can learn the protein representation efficiently and obtain good prediction effect. [Xie et al. \(2023\)](#) used two separate modules to automatically extract high-level semantic features and combined a learning strategy to deal with unbalanced data sets. [Yu et al. \(2024\)](#) developed a new integrated deep learning architecture, PhosAF, that considers sequence and three-dimensional structural information predicted by AlphaFold2. [Li et al. \(2022\)](#) utilized learning embedded features and attention architecture (adaptive framework) for prediction and verified a strong potential for application in the task of PTM site recognition. [Chang et al. \(2024\)](#) constructed DeepNphos and used them for the first time to predict N-phosphorylation sites. Its convolutional neural network (CNN) model combined with One-Hot encoding features showed good performance in predicting pK, pR and pH sites.

Inspired by this, an innovative mechanism called Res-GCN is introduced in this study to predict two kinds of protein phosphorylation sites. Firstly, eight feature extraction methods including AAindex, PseAAC, ANBPB, DC, BE, EAAC, Word2Vec, and BLOSUM62 matrices are utilized to digitize the protein sequence from multiple viewpoints, where the features obtained from different perspectives are fused in parallel into a complex feature vector. Secondly, the elastic net is utilized to determine out the influential subset of features in the fused matrix and remove the redundant information and irrelevant noise. Finally, the graph convolutional network (GCN) and residual network (ResNet) deep learning model is constructed to classify the phosphorylation sites. The studies demonstrate that the Res-GCN method is capable of identifying multiple phosphorylation post-translational sites compared to some other existing methods.

3. Materials and methods

3.1. Dataset

The collection of data is the one established by [Lv et al. \(2021a\)](#) and [Stukalov et al. \(2021\)](#) in the literature and is an experimentally validated phosphorylation site in SARS-CoV-2 infected human A549 cells. This dataset is more persuasive and makes it easier to compare the outcomes with those of other studies, enabling an objective evaluation of the models discussed in this study. To avoid the model being overfit and decrease the redundancy of protein sequences, the sequence homology threshold is set at 30 % using the CD-HIT algorithm ([Li and Godzik, 2006](#)). To facilitate a fair comparison with others' results, the truncated fragments of the processed sequences are all 33 in length and had S/T or Y in the middle. If a sequencing fragment contains a phosphorylated S/T or Y residue at position 17, it is classified as a positive sample; if not, it is classified as a negative sample. To maintain a consistent number of positive and negative samples, we randomly eliminate the non-redundant subset of negative samples. Finally, we get 102 positive and 102 negative examples for the Y sites, and 5387 positive and 5387 negative examples for the S/T sites. An 8:2 ([Qiao et al.,](#)

2022) random division of the processed samples is made into non-overlapping training and test sets. An exhaustive description of the above data is seen in Table 1.

3.2. Feature coding

3.2.1. Sequence information

3.2.1.1. BE. Binary encoding (BE) (Cui et al., 2019) is a widely used coding system that characterizes the relative positions and types of amino acids in the protein sequence. The way this coding technique operates by converting the processed protein sequence into a numerical matrix and feeding it into a classifier. For sequences ACDEFGHIKLMNPQRSTVWYX consisting of both 20 amino acids and pseudo-amino acids are transformed into corresponding 21-dimensional binary vectors. For example, residue E can be represented as (00010000000000000000). For the protein sequence of length N , the dimension of its final transformation into a numerical vector is $21 * N$.

3.2.1.2. ANBPB. The adapted normal distribution bi-profile Bayes (ANBPB) (Jia et al., 2017) algorithm models the data as normally distributed and uses Bayesian inference to estimate the unknown parameters. The random variable is independent and obeys a binomial distribution $b(n, p)$, $p = 1/20$ denotes the probability that each of the twenty amino acids will occur at a position, and n represents the twenty amino acids. When there is a big sample size and an independent chance of each amino acid occurring at each place, the central limit theorem is used to standardize the random variable x_{ij} , ij denotes the j -th position of the i -th amino acid, and thus approximate its distribution to the standard normal distribution obtained:

$$x'_{ij} = \frac{x_{ij} - np}{\sqrt{v_j}} \quad (1)$$

where v_j is the standard deviation. Then the posterior probability p_j is described as:

$$p_j = P(X \leq X_{ij}) = \xi(X'_{ij}) \quad (2)$$

where $\xi(x)$ obeys the standard normal distribution, $\xi(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^x e^{-\frac{t^2}{2}} dt$. For each protein sequence of length N , the dimension of its final transformation into a numerical vector is $2 * N$.

3.2.1.3. EAAC. Quantitative study of numerous amino acids in protein sequences using enhanced amino acid composition (EAAC) (Wang et al., 2020a), which characterizes the protein by determining the relationship between the proportion of each amino acid. Its calculation formula is:

$$f(t, win) = \frac{N(t, win)}{N(win)}, t \in (A, C, D, \dots, Y) \quad (3)$$

how many of type t amino acids in the window $win \in (window1, window2, \dots, windowN)$ is indicated by $N(t, win)$, and $N(win)$ is a sliding window with a size default value of 5. EAAC feature extraction results in a final 580-dimensional feature vector.

3.2.1.4. DC. An amino acid sequence with N residues in a protein can

have $N-1$ adjacent dipeptides. The periodicity of amino acids pairings for a particular sequence of proteins is calculated using the dipeptide composition (DC) (Wang et al., 2021). The formula is:

$$P_i = \sum_{i=1}^N C_i / (N-1), i = 1, 2, \dots, 400 \quad (4)$$

where C_i is the quantity of the i -th pair of amino acids that occurs, and N denotes the quantity of amino acids in each sequence. Finally, the DC feature extraction results in a 400-dimensional numerical vector.

3.2.2. Physical and chemical information

3.2.2.1. AAindex. The Amino acid property encodings (AAindex) database (Xiang et al., 2017) contains 566 indices, each with a corresponding set of values, which include biophysical, chemical, structural, and functional attributes of amino acids, etc. The best logical characteristics of biological reactions are the physical characteristics of amino acids, so if we choose all amino acid indices, not only can't reflect the physical-chemical properties around the post-translational modification sites effectively but also will generate redundant and noisy information. If all the amino acid indices are selected, not only can they not effectively capture the physicochemical properties surrounding the post-translational modification sites, but generate redundant and noisy data. As a result, 16 amino acid indices are selected in this paper. By utilizing the values corresponding to these 16 physicochemical properties, the sequences of phosphorylated and non-phosphorylated samples are transformed into numerical vectors. For samples with window size N , $16 * N$ -dimensional feature vectors are generated.

3.2.2.2. PseAAC. Pseudo-amino acid composition (PseAAC) (Malebary et al., 2021) introduces several additional pseudo-amino acids that are used to capture more information about the protein sequence. These pseudo-amino acids can be constructed based on amino acid properties, structure, or other features. The formula is:

$$m = [m_1, m_2, \dots, m_{20}, m_{20+1}, \dots, m_{20+\lambda}]^T (\lambda < N) \quad (5)$$

where the feature matrices $[m_1, m_2, \dots, m_{20}]$ and $[m_{20+1}, \dots, m_{20+\lambda}]$ represents amino acids composition information and location information in protein sequence, respectively. And N is the number of amino acids in a sample.

Each feature component in m is defined as:

$$p_t = \begin{cases} \frac{f_t}{\sum_{t=1}^{20} f_t + \omega \sum_{i=1}^{\lambda} \tau_i} & 1 \leq t \leq 20 \\ \frac{\omega \tau_{t-20}}{\sum_{t=1}^{20} f_t + \omega \sum_{i=1}^{\lambda} \tau_i} & 20+1 \leq t \leq 20+\lambda \end{cases} \quad (6)$$

the likelihood that amino acids of class t will appear in the sample is denoted by f_t , ω is the weight factor, and τ_i denotes the sequence correlation factor. The feature vector that PseAAC extracted has a dimension of $20 + \lambda$.

3.2.3. Evolutionary information

3.2.3.1. BLOSUM62. The BLOSUM62 matrix (Wang et al., 2022) is a commonly utilized protein sequence comparison tool that measures the probability of a pair of amino acids substitution. Considering these scores, protein sequence similarity and evolutionary relationships can be assessed. The BLOSUM62 matrix is based on comparative data between all amino acid patterns with no more than 60 % similarity. It is constructed based on statistical data and sequence similarity and has been extensively analyzed in protein databases. For samples with

Table 1
Sample size in the S/T and Y sites dataset.

Phosphorylation	Dataset	Positive	Negative
S/T	Training dataset	4308	4308
	Independent test dataset	1079	1079
Y	Training dataset	81	81
	Independent test dataset	21	21

window size N , $20 * N$ -dimensional feature vectors are generated.

3.2.4. Numerical space-derived information

3.2.4.1. Word2Vec. Word2Vec is an algorithm for representing words as vectors, which was proposed by Mikolov et al (Mikolov et al., 2013). in 2013. It can capture semantic information in protein sequences and map similar fragments to similar vector spaces (Guo et al., 2019). Word2Vec has two main models: the Continuous Bag of Words (CBOW) model and the Skip-gram model. The basic idea of both models is to learn target words or context to learn a word's vector representation. The CBOW model does this by summing and averaging the vectors of the context words and then mapping them to the target words through a hidden layer. The Skip-gram model works by passing the vectors of the target words to a hidden layer, which then maps them to the word vectors of the context. In terms of algorithm, the framework of the CBOW model reflects the benefit of a unified organization of distributed information. Therefore, the amino acid homology of samples is trained using the CBOW model in this work. Finally, the Word2Vec feature extraction results in a 200-dimensional feature vector.

3.3. Elastic net

Elastic net (Zou and Hastie, 2005) is a commonly used method for data reduction and feature selection. When dealing with high-dimensional data, it could better deal with redundant features and overfitting because of combining the advantages of Ridge Regression and Lasso Regression. The objective function of the elastic network consists of two parts: the first part is the squared addition of the Mean Square Error (MSE), and the second part is the weighted sum of the L1 regularization term and the L2 regularization term (Zhang et al., 2022). Its objective function is:

$$\min_w \frac{1}{2} \|y - Xw\|_2^2 + \alpha * \beta \|w\|_1 + \frac{1}{2} \alpha * (1 - \beta) \|w\|_2^2 \quad (7)$$

where X denotes the fusion matrix, y denotes the labels, n denotes the amount of samples, w means the weight coefficients, and α means the penalty parameter.

The L1 regularization term (Lasso regularization) tends to produce a sparse solution, i.e., the coefficients of some features are set to 0, thus enabling feature selection, while the L2 regularization term (ridge regression regularization) tends to shrink the coefficients of features in a balanced way. The elastic network balances the effects of feature selection and feature reduction by adjusting the weights between the L1 regularization term and the L2 regularization term. When the weight is 0, it is equivalent to ridge regression; when the weight is 1, it is equivalent to Lasso regression. In addition, the elastic network can effectively handle high-dimensional data, reduce the influence of redundant features, avoid the overfitting problem, and be able to perform feature selection and feature reduction simultaneously. In conclusion, by combining the advantages of L1 and L2 regularization terms, the elastic network can achieve better results in data reduction and feature selection.

3.4. Graph convolutional network

Graph convolutional network (GCN) (Feng et al., 2022; Kipf and Welling, 2017) is a deep learning model for graph data. GCN is similar to convolutional neural network (CNN), except that CNN is used for 2D data structure, GCN is by performing convolutional operation on graph structure, it can effectively capture the relationship between nodes and local structure, so as to realize feature learning and prediction on graph data. GCN is also a neural network layer that propagates from layer to layer:

$$H^{(l+1)} = \sigma(\tilde{D}^{-1/2} \tilde{A} \tilde{D}^{-1/2} H^{(l)} W^{(l)}) \quad (8)$$

In this formulation, $\tilde{A} = A + I$, I represents the unit matrix, $\tilde{N}$ represents the degree matrix of $\tilde{A}$, $H^{(l)} \in R^{N \times D}$ represents the characteristic of each layer, and σ represents the nonlinear activation function.

3.5. Residual network

Since deep learning has advanced, the number of layers of neural networks has been increasing to obtain better performance. However, deep neural networks face the issue of gradient vanishing and gradient exploding, which makes the network difficult to train (Han et al., 2022). In addition, information transfer in deep networks has become difficult, leading to a decrease in network performance. He et al. (2016) proposed a residual network (ResNet) in 2015 and achieved excellent results in the ImageNet image classification competition. The core idea of ResNet is the introduction of residual modules, which enables a deeper network structure by learning residuals instead of direct mapping. The basic unit of a residual network is the residual module. The residual module contains two or more convolutional layers and a skip connection. The skip connection adds the inputs directly to the outputs of the convolutional layers to form a residual connection. Let the input be x , and the output of the convolutional layers be $F(x)$, then the residual network formula is:

$$H(x) = F(x) + x \quad (9)$$

A residual network consists of multiple residual modules stacked together. Each residual module may contain within it a different number of convolutional layers, and different convolutional kernel sizes and step sizes.

3.6. Res-GCN architecture

Here, we propose a hybrid deep learning architecture, Res-GCN, consisting of GCN and ResNet. GCN can learn the relationship between neighbor nodes, and the ResNet solves the gradient disappearance problem in deep neural networks by introducing residual connections. The structure is as follows:

(1) GCN layer: It consists of input layer, hidden layer and output layer. The protein structure can be represented as a graph where each node represents an amino acid residue and the edges between the nodes represent the relationships between them. Firstly, knn_graph function is used to calculate the distance between nodes according to the input eigenmatrix, select the nearest neighbor node of each node, and construct the graph based on the nearest neighbor. Then the nearest-neighbor graph and the feature matrix are input into GCN, and the data are initially screened through the convolution of the two-layer graph.

(2) ResNet layer: The output of the upper layer network is fed into the residual network module, and each residual block learns the residual information by adding a residual connection, which helps the network better capture and learn low-level features in the input data. The residual module consists of a series of convolutional layers and activation functions, through which features are gradually extracted and these features are added to the original input to achieve information transfer and learning. Finally, the final feature representation is mapped to the result of binary classification through the fully connected layer.

This article uses Python version 3.6.5 and torch version 1.10.2 to train the proposed model. To train the model, we use the widely used K-fold cross-validation method, where k is tuned to 5. Dropout strategies are used to solve the problem of overfitting during training. In training, epoch, learning rate, and batch size are set to 30, 0.01, and 32, respectively. The optimizer is Adam and CrossEntropyLoss is used as a loss function.

3.7. Model evaluation

To assess how well the prediction model performs, we evaluate it using a 5-fold cross-validation and independent testing and utilize some widely-used metrics (Hu et al., 2022; Wei et al., 2022; Yu et al., 2022; Gao et al., 2023), including specificity (Sp), sensitivity (Sn), accuracy (ACC), and MCC. The formula is:

$$Sn = \frac{TP}{FN + TP} \quad (10)$$

$$Sp = \frac{TN}{TN + FP} \quad (11)$$

$$ACC = \frac{TN + TP}{TP + TN + FP + FN} \quad (12)$$

$$MCC = \frac{TN \times TP - FN \times FP}{\sqrt{(TN + FP)(TN + FN)(TP + FP)(TP + FN)}} \quad (13)$$

where TP and TN express the amount of accurately recognized phosphorylated and non-phosphorylated sites, and FP and FN express the amount of mistakenly recognized phosphorylated and non-phosphorylated sites, respectively. The ROC (Yin et al., 2022) and PR curves (Zhang et al., 2023) are also chosen to be the criteria for measuring the performance of the model, and AUC and AUPR are the

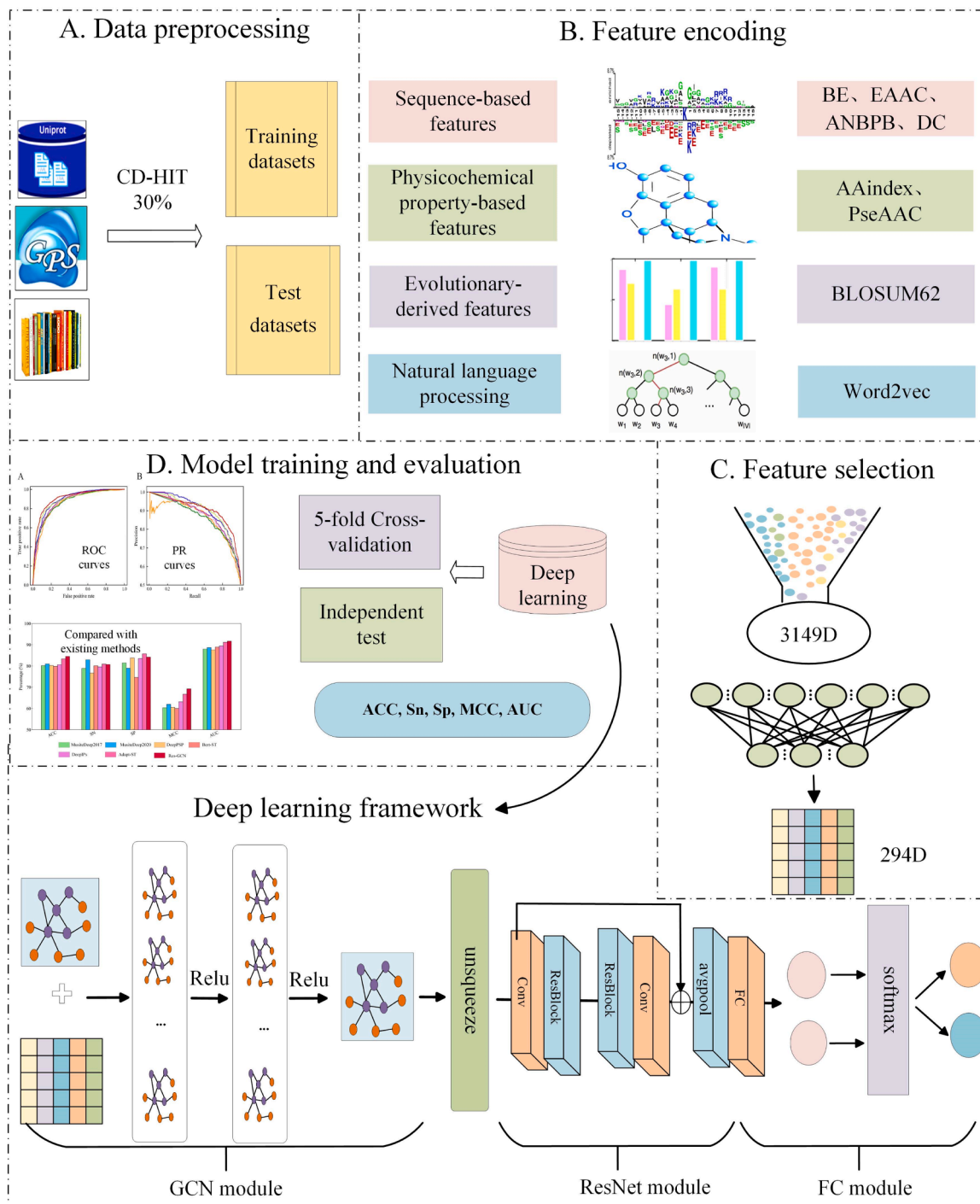


Fig. 1. Flowchart of the model Res-GCN.

size of the corresponding regions below the curves, respectively. The greater the area under the curves, the more accurate the prediction.

The significance of the difference between the two samples is tested by the two-sided T-test method (Wang et al., 2023) in the statistical test. The formula is:

$$t = \frac{\bar{E}_1 - \bar{E}_2}{\sqrt{\frac{(l_1-1)\delta_1 + (l_2-1)\delta_2}{l_1+l_2-2} \left(\frac{1}{l_1} + \frac{1}{l_2} \right)}} \quad (14)$$

where l_1 and l_2 are the two sample sizes, and the corresponding variance is δ_1 and δ_2 . The $\bar{E}_i$ denotes the mean of the two samples, respectively.

The phosphorylation site forecasting framework put forth in this paper is called Res-GCN, and the procedure is depicted in Fig. 1. The setting in which the experiment is conducted is: Windows Server 2012R2 Intel(R) Core (TM) i7-10510 U CPU @ 1.80 GHz 2.30 GHz and Python3.6. The datasets and source codes may be found at <https://github.com/YuBinLab-QUST/Res-GCN/>.

The following stages can be used to describe the Res-GCN model:

(1) Collect data. Two phosphorylation sites from the literature are collected and divided proportionally into the training samples and the testing samples.

(2) Feature coding. Eight feature encoding methods (AAindex, PseAAC, ANBPB, DC, BE, EAAC, Word2Vec, and BLOSUM62) are utilized to digitize the protein sequence from multiple viewpoints, which yields 528, 24, 64, 400, 693, 580, 200, and 660 dimensions, respectively, and multi-information fusion totaling 3149 dimensions.

(3) Feature selection. The influential features are obtained by using elastic net to remove the irrelevant or redundant features from the fusion features.

(4) Construct a classification model. GCN performs graph convolution operation on protein feature matrix and adjacency matrix, extracts feature and generates output. The GCN processed protein sequences are input into the residual network, through which deeper feature representations can be learned to achieve accurate prediction of modification sites.

(5) Model evaluation. Evaluate the model by selecting five-fold cross-validation and independent testing depending on the model's built-in steps (2)-(4). ACC, AUC, Sn, Sp, and MCC are taken as evaluation metrics.

(6) Model testing. To evaluate the capacity of Res-GCN for

generalization, the phosphorylated Y dataset is processed in accordance with the model developed in steps (2)-(4).

4. Results and discussion

4.1. Comparison of sample sequences

To analyze how frequently amino acids appear at each site flanking the phosphorylated and non-phosphorylated sites, Two-Sample-Logo (Vacic et al., 2006) is utilized to paint a double-sequence identification diagram of S/T and Y training sample fragments. The sequence length of the protein is 33, the description ranges from 1 to 33, and the sample sequences are analyzed as indicated in Fig. 2.

As demonstrated in Fig. 2, notable variations exist above the amino acid sequences surrounding phosphorylated and non-phosphorylated sites. On the S/T and Y samples, proline (P) and serine (S) are almost entirely concentrated in the upstream position, leucine (L) and lysine (K) are mainly enriched in the downstream position, and phosphorylation target sites, such as serine (S) and threonine (T), are enriched around the phosphorylation sites. For example, on the Y dataset, serine (S) is mainly enriched at positions 1, 2, 6, 8, 12, 13, 15, 16 and 20, 29, 32, glutamic acid (E) is mainly concentrated near positions 3, 13, 14 and 26, and leucine (L) is mainly enriched on positions 5, 9 and 25, 31 in the negative samples. Consequently, it is evident that the frequency of numerous amino acids occurring at varying points near phosphorylation varies considerably, highlighting the importance of different information in the protein sequence for feature coding.

4.2. Impact of multi-information fusion on results

In this research, eight feature encoder strategies are used to extract sequence characteristics, physicochemical characteristics, evolutionary information, and numerical spatial derived characteristics from protein sequences, and convert amino acid sequences into numerical representations. The feature matrices obtained from the eight methods are fused sequentially to obtain a 3149-dimensional fusion matrix, denoted by sum. The features obtained from each feature extraction method as well as the fused features are individually input into the classifier to obtain different metrics values, as shown in Table 2. The bolded data appearing in the table indicates the highest indicator value. In addition, to compare the prediction ability of the models more clearly, this paper plots the

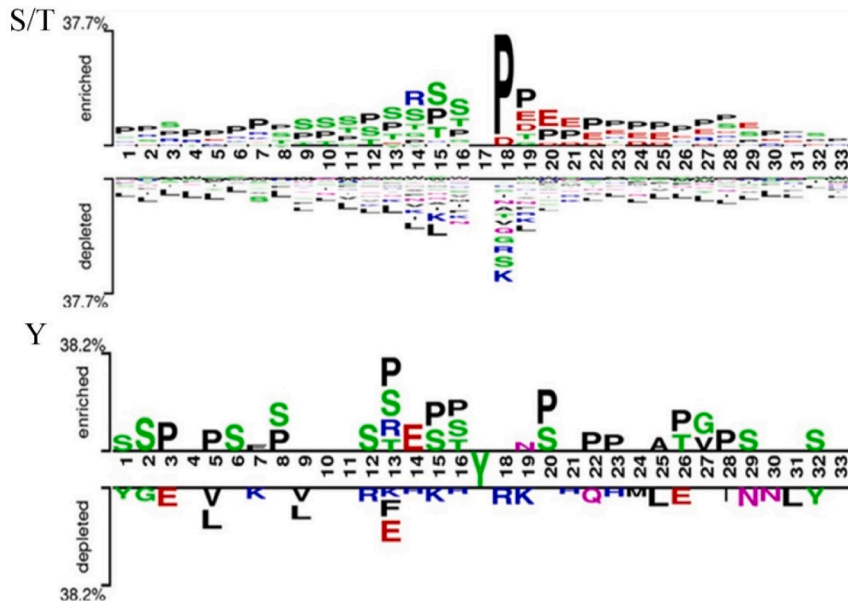


Fig. 2. amino acid distribution analysis of S/T and Y training datasets.

Table 2
comparing of results of single feature and integrated feature.

Method	ACC (%)	Sn (%)	Sp (%)	MCC
BE	66.63	68.41	64.84	0.3328
ANBPB	69.88	69.49	70.27	0.4075
EAAC	75.34	86.99	63.70	0.5212
DC	71.62	81.65	61.61	0.4415
AAindex	70.58	68.30	72.83	0.4244
PseAAC	70.87	79.21	62.53	0.4334
Word2Vec	72.44	73.64	71.23	0.4488
BLOSUM62	73.31	70.30	76.31	0.4670
sum	76.42	80.40	73.13	0.5282

ROC and PR drawing of the eight feature encoder strategies as well as the fused ones for the training set of S/T phosphorylation sites after 5-fold cross-validation, as seen in Fig. 3.

Table 2 illustrates the variations in the assessment metrics acquired for various feature extraction methods, and the accuracy rates of the eight feature extraction methods are 66.63 %, 69.88 %, 75.34 %, 71.62 %, 70.58 %, 70.87 %, 72.44 %, and 73.31 %, respectively. Among them, EAAC depended on sequence characteristics, Word2Vec depended on numerical space-derived characteristics, and BLOSUM62 based on evolutionary information have higher accuracy. Compared to alternative feature extraction techniques, the fused ACC and MCC metrics are much higher at 76.42 % and 0.5282, respectively. Among these, the MCC is greater than the comparable values of other approaches by 0.70 %-19.54 %, and the ACC metrics are higher by 1.08 %-9.79 % than other feature extraction techniques. Combining eight distinct feature extraction methods can yield more comprehensive and multi-angle information, as the table's various predictive signs demonstrate. The results of the evaluation indicators after feature fusion are all significantly better than the results of single features, which indicates that by fusing multiple feature extraction methods, the impact of the loss of information from single feature extraction on the model can be minimized, and the model's stability and robustness can be further enhanced.

As demonstrated in Fig. 3, the ROC and PR graphs after fusing data significantly exceed those of the eight single-feature data, which indicates that the fusion results in the largest AUC and AUPR percentages at 0.8453 and 0.8534. The percentages of AUC are superior to the other eight feature coding methods in the range of 1.10 %-12.52 %, and the AUPR values are superior to the other eight feature coding methods in the range of 2.06 %- 16.21 %. The comparison shows that the fused feature extraction algorithm gets good results, which is because feature

fusion can describe the data from different perspectives, and a more comprehensive and diversified data representation can be obtained through fusion, which improves the precision of the model.

To explore the effect of each feature encoding on the model performance, we test the model performance in turn without AAindex, PseAAC, ANBPB, DC, BE, EAAC, Word2Vec, and BLOSUM62. The experimental results are shown in Fig. 4. In all model results with feature coding removed, Sn, Sp, ACC and MCC all showed varying degrees of decline. Among them, MCC decreased by at least 2.42 % and ACC decreased by at least 1.77 %. The results show that the eight feature coding methods are the key parts of the model, which can extract protein sequence information from different aspects.

4.3. Impact of different feature selection on results

To find the optimum subset of features after feature fusion, irrelevant or redundant features are eliminated to reduce the dimension of the feature matrix. In this article, we choose seven feature selection methods, Group Lasso (Yuan and Lin, 2006), Lasso (Tibshirani, 1996), MI (Ross, 2014), IG (Chen et al., 2007), GINI (Jiang et al., 2020), FE (Wang et al., 2020b), and elastic net (Zou and Hastie, 2005; Zhang et al., 2022) to characterize the data from the fused feature vectors, and the seven feature selection methods are sequentially processed the S/T training dataset into 721, 470, 370, 178, 200, 842, and 294-dimensional datasets. To compare the superiority of the feature selection methods, we used the five-fold cross-validation, the corresponding ROC and PR graphs are also plotted, these can be found in Fig. S1, and Table 3 displays the specific examination results.

Table 3 shows that the variations in the assessment metrics acquired for seven feature selection methods, and the AUC, ACC, and MCC values obtained by the elastic net method are 90.82 %, 84.38 %, and 0.6875, respectively, in which the ACC values are increased by 2.03 %-8.28 %, and the AUC values are increased by 1.10 %-4.67 %, which are all superior to the other feature selection methods. The accuracy of the data processed by elastic net is obviously improved more than that of unprocessed data, which indicates that the elastic net strategy filters out the values that do not contribute much to the model through regularization, effectively reduces the data dimension, selects better features, and improves the efficiency and classification performance of the model.

4.4. Feature visualization

A visualization technique called t-SNE (Maaten and Hinton, 2008)

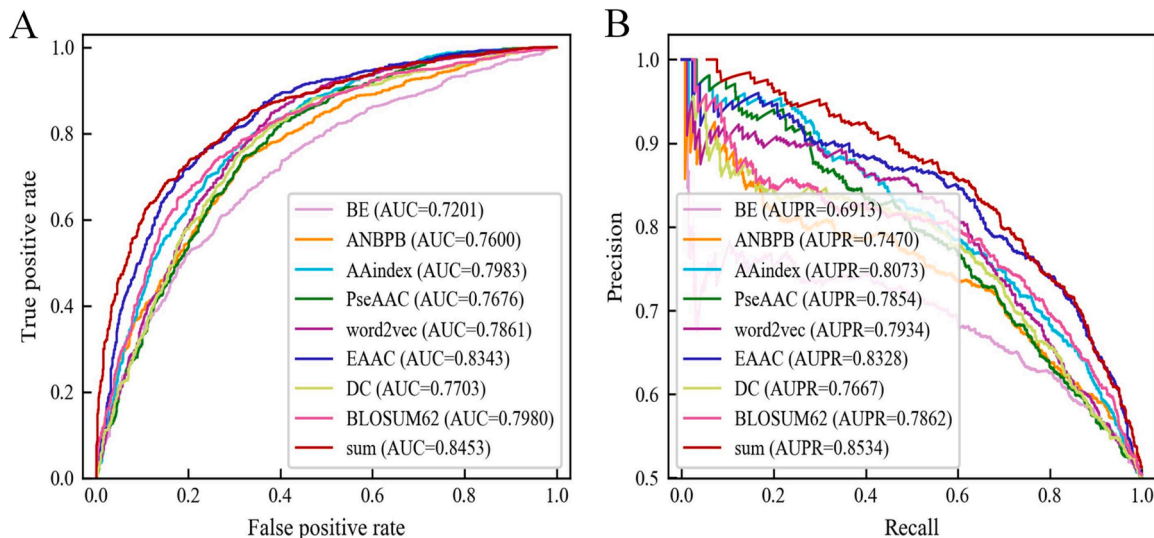


Fig. 3. ROC and PR curves of eight feature encoder strategies on S/T training set (A) ROC curves (B) PR curves.

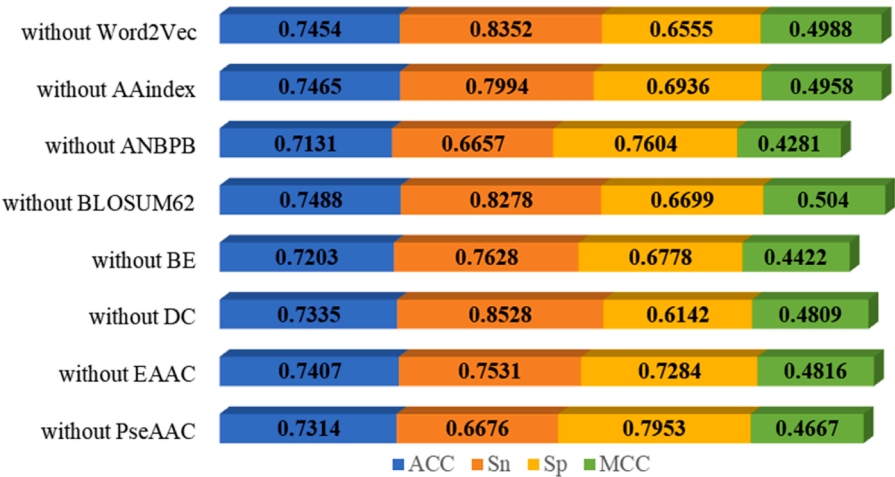


Fig. 4. Performance comparison of any seven feature encodings.

Table 3
comparing of results of seven feature selection algorithms.

Method	ACC (%)	Sn (%)	Sp (%)	MCC	AUC
Group Lasso	82.15	81.83	83.14	0.6426	0.8898
Lasso	82.35	83.99	80.72	0.6474	0.8972
MI	80.49	81.32	79.68	0.6101	0.8789
IG	78.82	73.77	82.96	0.5598	0.8641
GINI	76.10	64.97	87.22	0.5353	0.8615
FE	80.13	79.02	81.24	0.6028	0.8711
elastic net	84.38	84.16	84.58	0.6875	0.9082

aims to make it simple to discern between positive and negative samples by placing comparable data points from high-dimensional data near to one another in a low-dimensional environment. In this study, we show the protein sequence distribution of the S/T dataset before and after the elastic net strategy using t-SNE. For the purpose of visualizing the form and trend of positive and negative datasets in the data more intuitively, we plot a scatter plot of the protein data. Fig. 5 displays the results of the visualization.

It is clear from Fig. 5, that in the S/T dataset, the positive and negative sample sequences are stacked together chaotically before elastic net processing, and the sites of phosphorylation and non-phosphorylation could not be clearly distinguished from each other. After processing the data by elastic net method, it can be clearly seen that the different categories have been clearly differentiated. The

negative samples are mainly aggregated at the bottom and the positive samples are basically aggregated at the top, which means that the processed data can distinguish the arrangement of the two types of samples more effectively.

4.5. Impact of classification algorithms on results

In this research, nine classifier models are chosen for assessment, including Bagging (Breiman, 1996), LightGBM (Liu et al., 2020), NB (Friedman et al., 1997), GTB (Ai et al., 2017), CNN (Wen et al., 2021; Chen et al., 2021), RNN (Cho et al., 2014), LSTM (Graves and Schmidhuber, 2005), and GCN (Feng et al., 2022; Kipf and Welling, 2017), in which the default parameters of Bagging, LightGBM, NB, and GTB are chosen. Table S1-S4 lists the parameters of CNN, RNN, LSTM, and GCN. After feature selection, we input the feature vectors into nine classifiers, and evaluate the model Res-GCN using 5-fold cross-validation. Table 4 provides the specific evaluation results of various classifiers on the S/T training samples. Furthermore, to better confirm the performance of the Res-GCN, we test each of the nine types of classifiers using a two-sided T-test (Wang et al., 2023). At the significance level, we compare the P-values of Bagging, CNN, LightGBM, NB, GTB, LSTM, and GCN on ACC, MCC, and AUC metrics with those of the Res-GCN classifier. Supplementary Table S5 displays the results of the statistical test.

From Table 4, the nine classifier models we have selected get different prediction accuracies. The corresponding ACC values of the nine classifier models are 77.67 %, 82.95 %, 76.99 %, 82.20 %, 77.67 %, 82.95 %, 76.99 %, 82.20 %, 77.67 %.

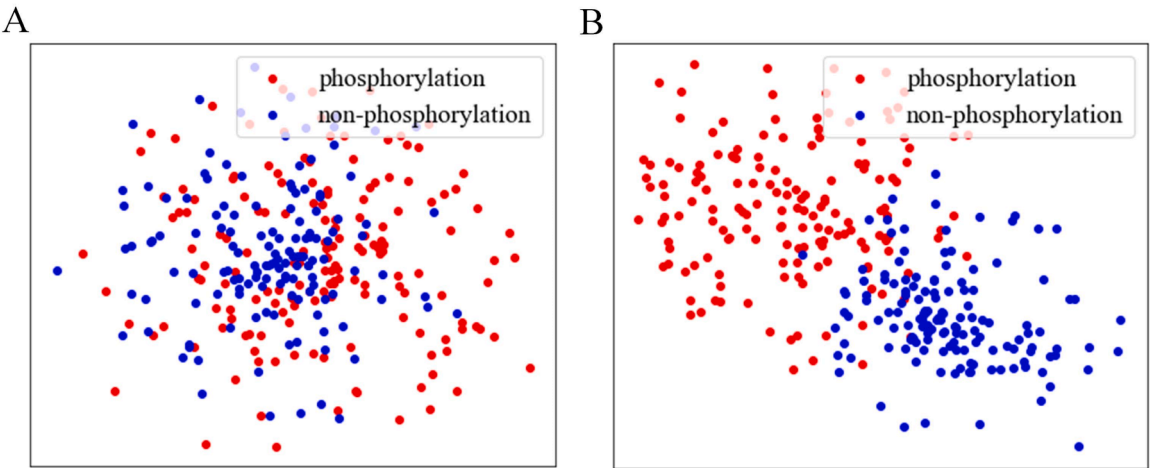


Fig. 5. Data visualization before and after elastic net process on S/T dataset.

Table 4
comparing of results of nine classifier algorithms.

Classifier	ACC (%)	Sn (%)	Sp (%)	MCC	AUC
Bagging	77.67	74.09	81.24	0.5549	0.8484
LightGBM	82.95	83.26	82.64	0.6591	0.9097
NB	76.99	82.78	71.21	0.5438	0.8413
GTB	82.20	82.22	82.19	0.6442	0.9008
CNN	79.99	80.15	79.83	0.6001	0.8788
RNN	80.82	80.34	81.31	0.6166	0.8852
LSTM	82.21	83.63	80.80	0.6448	0.9027
GCN	83.34	83.72	82.96	0.6671	0.8730
Res-GCN	84.38	84.16	84.58	0.6875	0.9082

79.99 %, 80.82 %, 82.21 %, 83.34 %, and 84.38 %, respectively. When compared to the other eight classifier models, the model Res-GCN developed in this work has the best prediction accuracy. The percentages of Sn, Sp, MCC, and AUC are basically higher than the other eight classifiers, in which the MCC values are greater than the comparable values of other classifiers by 2.04 %-14.37 %, the ACC is greater by 1.04 %-7.39 %. Therefore, the model Res-GCN constructed in this paper is adopted as the most effective classification model for phosphorylation site prediction.

In this paper, we also plot the boxplot (Yin et al., 2022) of the nine classifier algorithms with respect to the AUC values, as depicted in Fig. 6. From the figure, for the S/T training samples, the corresponding AUC percentages of different classifier algorithms have large differences. Among them, the four classifier algorithms LightGBM, GTB, and Res-GCN have AUC values above 0.9, but the overall data volatility of the AUC obtained from the model Res-GCN constructed in this paper is small and there are no outliers. The classifier algorithms LightGBM and GTB each have points located outside the box, which are anomalies. By plotting the box line diagram, the average level of the data, the degree of volatility, and outliers can be identified intuitively and clearly, and the results illustrate the strong predictability and steadiness of the Res-GCN model developed in this work.

4.6. Comparison of advanced models

To illustrate the validity of the Res-GCN model more fairly, it is compared with the extant models MusiteDeep2017 (Wang et al., 2017), MusiteDeep2020 (Wang et al., 2020c), DeepPSP (Guo et al., 2021), Bert-ST (Qiao et al., 2022), DeepIPs (Lv et al., 2021b), Adapt-ST (Li et al., 2022) and PSPred-ALE (Jiao et al., 2023) in the same dataset models for comparison. A comparison of the evaluated values for each of the phosphorylated S/T independent test datasets is seen in Table 5. To compare the contrasting results of various models for prediction, bar graphs are plotted more visually and displayed in Fig. S2.

From Table 5, for the phosphorylation S/T independent test dataset,

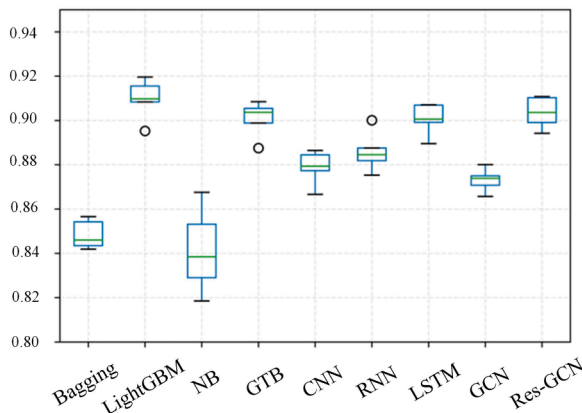


Fig. 6. Boxplot of AUC values for different classifier algorithms.

Table 5
Comparison of results of existing model on S/T testing dataset.

Model	ACC (%)	Sn (%)	Sp (%)	MCC (%)	AUC (%)
MusiteDeep2017	80.17	78.87	81.46	60.35	87.98
MusiteDeep2020	80.95	82.95	78.96	61.96	88.67
DeepPSP	80.21	76.65	83.78	60.58	87.62
Bert-ST	79.84	80.07	74.60	60.00	88.90
DeepIPs	80.63	79.61	83.50	63.16	89.37
Adapt-ST	83.32	80.90	85.72	66.70	91.20
PSPred-ALE	83.06	82.58	83.54	66.10	90.70
Res-GCN	83.50	79.24	87.76	67.25	91.32

the ACC, MCC, and AUC percentage of the model Res-GCN are 83.50 %, 67.25 %, and 91.32 %, respectively, which are greater than the corresponding indexes of other methods. Among them, the ACC values are higher than those of other models by 0.18 %-3.66 %, the MCC values are higher by 0.55 %-7.25 %, the AUC values are higher by 0.12 %-3.70 %. Sp also achieve better results, which fully demonstrate that the model Res-GCN can effectively predict the phosphorylation sites, especially after removing the redundant data by using the feature selection method, the predictive ability of the Res-GCN is greatly improved.

4.7. Testing model generalization capabilities

The model Res-GCN is selected to be retrained using the phosphorylated Y dataset, after which it is tested using an independent test to further assess the model's capacity for generalization. We compare using the same dataset with the extant models MusiteDeep2017 (Wang et al., 2017), MusiteDeep2020 (Wang et al., 2020c), DeepPSP (Guo et al., 2021), and DeepIPs (Lv et al., 2021b). Table 6 displays a comparison of each assessment value in the Y-independent test dataset. To contrast the results of various models more visually, bar charts are plotted in this paper, as shown in Fig. S3.

From Table 6, for the phosphorylation Y independent test dataset, the ACC, MCC, and AUC percentage of the model Res-GCN are 88.10 %, 0.7628 and 0.9773, respectively, which are greater than the corresponding indexes of other methods. Among them, the ACC values are higher than those of other models by 2.59 %-11.91 %, the MCC values are higher by 3.52 %-19.63 %, the AUC values are higher by 5.21 %-16.32 %. Sn and Sp also achieve better results, and the difference between Sn and Sp values is small compared with other methods. The different index results fully indicate that the model Res-GCN makes appropriate predictions for phosphorylated Y sites, reflecting the good generalization ability of the model.

4.8. Interpretable analysis

It is demonstrated through studies that the Res-GCN model presented in this research is capable of accurately predicting the phosphorylation sites on proteins, but it is also necessary to understand the idea behind the model architecture and its biological significance. The motif is a typical sequence or a structure, sequence motifs are becoming more and more important in bioinformatic analyses. It is possible to extract protein sequences from the matching motif, and if the predicted protein sequence has excellent sequence similarity to a given protein sequence, it is considered to be the correct protein phosphorylation modification

Table 6
Comparison of results of existing model on Y testing dataset.

Model	ACC (%)	Sn (%)	Sp (%)	MCC (%)	AUC (%)
MusiteDeep2017	80.95	85.71	76.19	62.19	81.41
MusiteDeep2020	85.51	95.24	76.19	72.76	87.30
DeepPSP	76.19	95.24	57.14	56.65	82.09
DeepIPs	83.33	90.48	80.95	71.75	92.52
Res-GCN	88.10	90.45	85.71	76.28	97.73

site. Given the Res-GCN model's strong predictive performance, it is assumed that the model can learn protein sequence information, which is a key feature for distinguishing phosphorylation modification sites from non-phosphorylation modification sites, where the model trained on the phosphorylated Y dataset is used as an example. First, the motifs recognized by the model Res-GCN were analyzed using MEME 5.5.0, and then the above motifs were analyzed for similarity with known motif databases using the tool Tomtom (He et al., 2023).

The motif logo consists of a bunch of letters at each position, as shown in Fig. 7, there are four plots ABCD, in each plot the first row is the motifs obtained from the known motif database and the second row is the motifs identified from the model Res-GCN. By finding similar motifs for analysis and comparison, it is evident that the motifs derived from these parts enable the framework to correctly determine protein PTM sites as phosphorylation sites, and these motifs are very similar to those of the known motif database. Among them, the A-D matched Tomtom p-values of $1.49\text{e-}05$, $8.80\text{e-}06$, $1.28\text{e-}04$, and $1.40\text{e-}04$ in Fig. 7 represent the most similar motifs in this study, which are statistically significant ($p\text{-value} < 1\text{e-}03$). Thus, it can be seen that the model is able to effectively learn distinct motifs. In addition, by increasing the transparency of the model in this way, it is shown that deep learning applied to protein sequences is biologically interpretable. The motifs characterized protein phosphorylation sites that may have specific functional roles in the cell, which will promote to further comprehend the function and machine processed of action of proteins and provide new ideas for biomedical research.

5. Conclusion

Phosphorylation is the most prevalent and extensively researched protein post-translational modification (PTM), which is essential to several biological functions. Since experimental validation of sites is limited by time and money, the development of computer techniques for precise prediction of phosphorylation sites has received a lot of attention. In this paper, we provide a new protein phosphorylation site identifying model, Res-GCN, which combines graph convolutional networks and residual networks for the first time, and uses residual connections to reduce feature loss. In order to verify the validity of the model, we have carried out a large number of verification experiments. Experiments show that the model put out in this work performs better than the models currently in use for predicting protein phosphorylation sites, especially for small training data. In addition to being excellent at site recognition, our model is also efficient at learning obvious motifs. These motifs characterize protein phosphorylation sites, which will contribute to further understanding of protein function and mechanism of action.

Although the model Res-GCN constructed in this work has substantially boosted accuracy of phosphorylation sites, there is still a need for improvement. The Word2Vec is limited to a fixed representation of word vectors and is unable to capture the various ways that words might have meanings in various situations. In the future, we will delve into dynamic word vector models such as ELMo, BERT, etc., which represent each word as a dynamic word vector using pre-training and fine-tuning, so that the model has a stronger word representation. It is also planned to

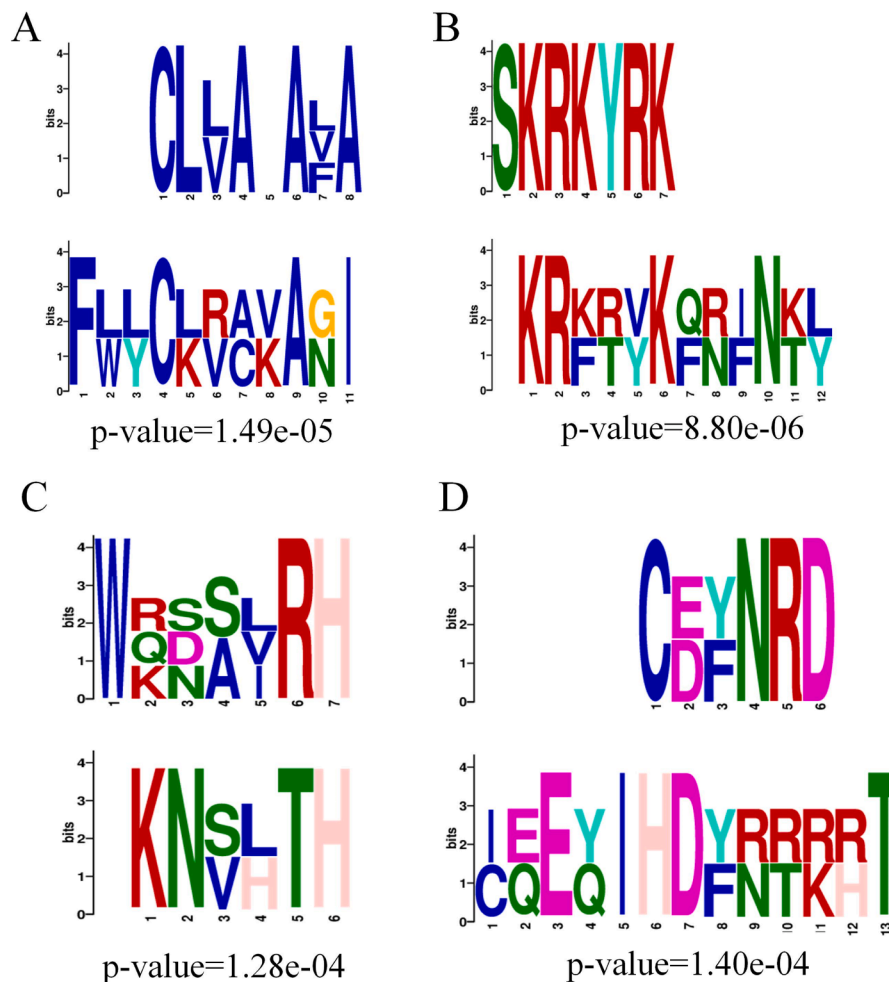


Fig. 7. Interpretability of the model. (A-D) Comparison of the database and the four pairs of motifs identified by the model as the best matches, with the lowest p-value (lower is better) for each match shown at the bottom.

construct a more detailed dataset, including data from different sites of multiple species, to offer more insightful direction for experimental studies of protein PTM sites.

CRedit authorship contribution statement

Jihua Jia: Writing – original draft, Visualization, Validation, Software, Methodology, Formal analysis, Data curation. **Minghui Wang:** Writing – original draft, Visualization, Validation, Supervision, Methodology, Formal analysis. **Bin Yu:** Writing – review & editing, Writing – original draft, Validation, Supervision, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Conceptualization. **Yushuang Liu:** Writing – review & editing, Writing – original draft, Visualization, Validation, Methodology, Investigation, Formal analysis. **Hongyan Zhou:** Validation, Methodology, Formal analysis, Data curation. **Fei Xu:** Visualization, Validation, Investigation, Formal analysis.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at [doi:10.1016/j.compbiolchem.2024.108183](https://doi.org/10.1016/j.compbiolchem.2024.108183).

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